

FREEDOM INTERNATIONAL SCHOOL
COMPUTER SCIENCE 2026-27
Binary File Worksheet

CS Board 2024 Main Paper

- i) Differentiate between 'w' and 'a' file modes in Python.
 - ii) Consider a binary file, items.dat, containing records stored in the given format:
{item_id: [item_name, amount]}
- Write a function, Copy_new(), that copies all records whose amount is greater than 1000 from items.dat to new_items.dat.

OR

- i) What is the advantage of using with clause while opening a file in Python? Also give syntax for with open.
- ii) A binary file Emp.dat has the following structure: [Emp_Id, Name, Salary]. Write a user defined function, disp_Detail(), that would read the contents of the file Emp.dat and display the details of those employees whose salary is below 25000.

CS Board 2024 Sample Paper

- i) Differentiate between r+ and w+ file modes in Python.
- ii) Consider a file Sports.dat, containing records of the following structure:
[SportName, TeamName, No_of_Players]. Write a function, copyData(), that reads contents from the file Sports.dat and copies the records with sport name as 'Basketball' to the file named 'Basket.dat'. The function should return the total number of records copied.

OR

- i) How are the text files different from the binary files?
- ii) A binary file CINEMA.DAT has the following structure: {MNO: [MNAME, MTYPE]}. Write a user defined function, findtype(mtype), that accepts mtype as a parameter and displays all those records from the file that have value of MTYPE as mtype.

CS Board 2024 Sahodva Paper

- i) Differentiate between 'w' and 'a' file modes in Python.
- ii) Consider a file CINEMA.DAT, containing records of the following structure:
[MovieName, Movie_Id, No_of_Theatres]. Write a function copy(), that reads contents from the file from the file CINEMA.DAT and copies the records with the movie names as 'KAALAI' to the file named CINEMA1.DAT. The function should return the total number of records copied to the file CINEMA1.DAT.

OR

- i) Differentiate between seek() and tell() in Python.
- ii) A binary file, CINEMA.DAT has the following structure {MNO:[MNAME, MTYPE]} where MNO- Movie number, MNAME- Movie name and MTYPE is Movie type. Write a user defined function, findtype(mtype), that accepts mtype as

a parameter and displays all the records from the binary file CINEMA.DAT, that have the value of movie type as mtype.

CS Board 2024 Supplementary Paper

- i) What is the main purpose of seek () and tell () method?
- ii) Consider a binary file, Cinema.dat containing information in the following structure: [Mno, Mname, Mtype].
Write a function, search_copy (), that reads the content from the file Cinema.dat and copies all the details of the "Comedy" movie type to file named movie.dat.

OR

- i) Give one difference between write() and writeline() function in text file.
- ii) A Binary file, "Items.dat" has the following structure: [Icode, Description, Price]
Where Icode – Item code Description – Detail of item Price – Price of item. Write a function Add_data (), that takes Icode, Description and Price from the user and writes the information in the binary file "Items.dat".

CS Board 2025 MainPaper

A file passenger.dat, stores the records of the passengers using the following structure: [PNR, PNAME, BRDSTN, DEST, FARE]. Write the user defined functions for the following:

- a) Create (-) to input data for passengers and write it in the binary file.
- b) Search(D)- to read contents from the binary file and display details of those passengers whose DEST matches with D.
- c) Updatefare (-) to increase the fare of all passengers by 5% and rewrite the updated records into the file.

CS Board 2025 Sample Paper

Surya is a manager working in a recruitment agency. He needs to manage the records of various candidates. For this, he wants the following information of each candidate to be stored: -

Candidate_ID – integer - Candidate_Name – string - Designation – string - Experience – float
You, as a programmer of the company, have been assigned to do this job for Surya.

- a) Write a function to input the data of a candidate and append it in the binary file.
- b) Write a function to update the data of candidates whose experience is more than 10 years and change their designation to "Senior Manager".
- c) Write a function to read the data from the binary file and display the data of all those candidates who are not "Senior Manager".

CS Board 2025 Sahodya Paper

Suresh is a manager working in a Company. He needs to store, update and display the records of sales by Salesperson for his company. For this, he wants the following information of each record to be stored: -

SalesPerson_ID – integer - Salesperson_Name – string - Product_Sold – string - Quantity – float

The format of each record is [SalesPerson_ID, Salesperson_Name, Product_Sold, Quantity]

- a) AddRecord() : He needs to input the data of a salesperson and add it in a binary file “record.bin”.
- b) UpdateFile() : he has to update a record based on the SalesPerson_ID entered by the user and also enter the updated Quantity. The updated record is to be written in the new file extra.bin. The records which are not to be updated also have to be written to the file extra.bin. If the SalesPerson_ID is not found, an appropriate message should be displayed.
- c) DisplayFile(sid): He has to read the data from the binary file extra.bin and display the data of that salesperson whose Salesperson_ID matches with the argument passed in the function.

OR

Deepak is a Python programmer working for election commission, For the upcoming election he has created a binary file “Election.bin”. The structure of Election.bin is:

[Party_Id, Party_Name, Candidate_Name, TotalVote]

Where Party_Id is Party ID (integer), Party_name is Party Name (string), Candidate_Name is name of candidate(string), TotalVote is total vote received by the candidate

For efficiently maintaining data of the event, Deepak wants to write the following user defined functions:

- a) accept() – to accept a record from the user and add it to the file Election.bin
- b) Search(PartyName) – to display the record for a particular person name and send the appropriate message if such record does not exist.
- c) update() – to update the Candidate name and Total votes based on the Party_ID. As a Python expert, help him complete the task.

CS Board 2025 Supplementary Paper

Keshav is the IT Head in a hospital. He needs to manage the records of all the doctors in the hospital. For this, he wants to store the following information of each doctor in a file :

D_ID – An integer to store Doctor ID.

D_Name – A string to store doctor’s name.

D_Dept – A string to store the Department of the doctor. (Surgery, Radiology, etc.)

Experience – An integer to store doctor’s experience (in years)

For example, a doctor’s information may be : [1256, 'R. Gupta', 'Cardiology', 15]

As an applicant for the post of a Programmer, you have to answer the following questions in this context :

- a) Write one difference of storing this data in a binary file over a CSV file.
- b) Assume that the data is stored in a binary file, named DOCTORS.DAT, and each record is stored as a list. Write a function, in Python, to read and display all the records from the file DOCTORS.DAT.
- c) Write a function addDoctor(), in Python, which accepts a doctor’s data from the user and writes it in the file DOCTORS.DAT.

CS Board 2026 Main Paper

NextStep is an organization which has a pool of resource persons to conduct training workshops on various topics related to ICT. The data of all its Resource Persons is stored in a binary file RESOURCES.DAT using the following record structure (each record is a tuple) :

(R_ID, R_Name, R_Expertise, Charges)

where : R_ID – Resource Person's ID (An integer)

R_Name - Resource Person's Name (A string)

R_Expertise - Area of expertise of the Resource Person

Charges - Charges (in rupees) per hour to conduct a workshop

For example, a record in the file is : (12, 'P. Velusami','Machine Learning',5000) In this context, write the following user defined functions in Python :

- a) Append() – To input the data of a Resource Person and write it in the file RESOURCES.DAT.
- b) Update() – To increase the Charges of each resource person by 500.

CS Board 2026 Sample Paper

Mr. Ravi, a manager at a tech company, needs to maintain records of employees. Each record should include: Employee_ID, Employee_Name, Department and Salary. Write the Python functions to:

- a) Input employee data and append it to a binary file.
- b) Update the salary of employees in the "IT" department to 200000.

CS Board 2026 Sahodva Paper

Vaibhavi is studying in Class 12 and trying to develop a solution for an easy record handling of student's result and performance. She wants to store the following information of the students in the form of a list: AdmNo-admission number, name, percentage. Her teacher guided her to use binary files for this purpose and create the following user defined functions:

- a) ADD_DATA()- to input the records and add it to the binary file.
- b) CountRec()- Accepts a range of percentage, display the records and count of the records lying in the given range.

Practice Questions

1. Mention any two differences between seek () and tell (). Consider a file FLIGHT.DAT containing multiple records. The structure of each record is as shown below: [Fno, Fname, Fare, Source, destination]
Write a function COPY_REC () in Python that copies all the records from the file FLIGHT.DAT where the source is DELHI and the destination is MUMBAI, into a new file RECORD.DAT.
2. Mention any two differences between binary and csv files. Consider a binary file BOOK.DAT containing a dictionary having multiple elements. Each element is in the form {BNo: [BName, BType, Price]}. Write a user defined function findBook(price), that accepts price as a parameter and displays all those records from the binary file BOOK.DAT which has a book price more than or equal to the price value passed as a parameter.
3. Consider a binary file STOCK.DAT that has the following structure: [OrderId, MedicineName, Qty, Price] of all medicines of Wellness Medicos. Write the following functions:
 - a) AddOrder ()- that accepts details of a medicine from the user and add it to the file.
 - b) Display ()- to display details of all medicine that have price more than 500.

4. Raj as a supervisor at a software development company. He needs to manage the records of various employees. For this, he wants the following information of each employee: [EmpId, Name, Designation, Salary]. You as a programmer, have been assigned to do this job for Raj.
 - a) Write a function to input data of an employee and append it to the binary file.
 - b) Write a function to update the data of employees whose salary is greater than 50000 and change the position to 'Team Lead'.
 - c) Write a function to read data from the binary file and display the data of those employees who are not 'Team Lead'.
5. Ms. Joshika is the lab attendant of a school. She is asked to maintain the project details of the project synopsis submitted by the students for the upcoming Board Exams. The information is: {Proj_Id: [Proj_Name, Members, Durations]}. As a programmer for the school, you have been asked to do this job for her by defining the following functions:
 - a) Prj_Input()- to input data of a project of students and append it to the binary file.
 - b) Prj_update()- to update the project details(duration as 4 months) whose member are more than 3.
6. Which file can be opened with notepad as well as MS Excel? A binary file "Garment.dat" exists containing records of different types of garments as per following structure: [GID, Type, Gender, Cost]. Write a Python program to add and search records of garments from the binary file and display using the following functions:
 - a) Addgarment () – function to input details of garments and store them to the file "Garment.dat", if the garment type is "cotton" or "silk"
 - b) Showgarment() – To read and display the records of "Silk" type and also display the count of the records displayed.
7. "KheloSports Pvt Ltd" is a sports management company that maintains a digital database of players in various sports. The company uses a binary file named "PLAYER.DAT" to store player information efficiently. The file contains records in the given format: {player_id: [pname, age]}. Write the user-defined functions in Python to perform the following actions:
 - a) Add_Data (): Take data of 5 players and write it to the binary file.
 - b) Delete_Record (): Remove the records of those players whose age is less than 18 years from the file "PLAYER.DAT".
8. Subha is a book shop owner. She needs to manage the records of various books in her shop. For this she wants the following information of each candidate to be stored: [Book_ID, Book_Name, Author, Edition]. You, as a programmer of the company, have been assigned to do this job for Subha.
 - a) Write a function to input the data of a book and append it to a binary file.
 - b) Write a function to update the records of books where the edition number is less than 5 and increase the edition number by 1.
 - c) Write a function to read the binary file and display the data of all those books whose author is "Sam".